CS 475 Scott Barlowe
Spring 2026

FINAL PROJECT AND PRESENTATION

OVERVIEW

The final project consists of a code artifact with demonstrated functionality and a presentation. The project proposal is due at 5:00 pm on March 27. Final projects will be due at 11:15 am on April 24. ONE person should submit your slides and code to Canvas. EACH person should submit an attestation by that time as well.

EACH person should make sure to redeem at least one coupon if they have not already.

GRADING

Your final project grade will consist of 70% for the coding part, 25% for the presentation, and 5% for the proposal. Your project will be graded on both design and functionality. Your presentation will be graded on content, length, and professionalism.

INDIVIDUAL CONTRIBUTIONS

Each team member must submit a detailed description of their contributions to the code and presentation. Individual or group grades may be altered based on the quality of the attestation, contributions to the code, and contributions to the presentation. **EACH PERSON MUST MAKE A SUBSTANTIAL CONTRIBUTION TO THE CODE.**

The instructor reserves the right to ask for demonstration/explanation/exam of any team member for any part of the assignment. The instructor reserves the right to alter the assignment grade of any individual based on an assessment of the amount and quality of contribution (in either the code or presentation).

CODE PRODUCT AND FUNCTIONALITY

Projects should

- Consist of a minimum of 400 lines of code not counting white space and documentation.
- Not be extensions of projects already completed.
- Rely on GCP for at least part of the code and functionality.
- Not use Generative AI to produce the code for which you plan to receive credit.
- Adhere to accepted documentation, coding conventions, and style.

SOME ADVICE

A successful project for this course will have a well-defined scope, can be completed within the time allowed, and provide enough depth that you gain valuable experience. Some components that you should consider include:

1. **The overall aim of the project.** What will the final product be able to do? What function/purpose will your project serve? Why is this function useful? What data set will you be using.
2. **The parts of GCP on which your project will rely**. Other than the most fundamental parts of GCP, what tools, toolkits, and APIs do you think will be necessary? Why are each of those tools, toolkits, and APIs necessary for your project?
3. **How much will utilizing GCP cost**? You need to do your best to estimate the cost so that you have enough credits on your coupon to complete your project. Research the data, resources, and time.

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4. **Monitor your GCP billing CLOSELY.** If you need additional coupons, you MUST allow time for your instructor to contact Google. This does not guarantee that additional coupons may be available.
5. **How much original code that you (or your team) will need to write.** How many lines of code (excluding comments and blank lines) do you estimate will need to be written by your team? What parts will require the most code to be written? Why do you think those parts will take more code than the other parts?
6. **Your team structure and member responsibilities.** How do you envision organizing your team? For what specific aspect of the project will each team member be responsible?
7. **The progression of your project.** How will you order the steps necessary for project completion? Why will you complete the project in that order?
8. **Which completed labs (or labs to be completed) do you envision will be the most relevant/helpful to your project.** Why do you think those are the most relevant/helpful?
9. **If you anticipate needing to learn anything outside of the skills acquired in the completed labs (or labs to be completed).** If so, what are the skills that you will need to acquire and how will you acquire them?

PRESENTATION

Presentations should

- Be 15 minutes in length and have the speaking roles for team members distributed equally. Each team member must have a significant speaking role equal to the other members.
- Describe the main aim of your project and why the problem solution is suitable for cloud-computing.
- Provide a detailed description of each cloud component in your solution, the role of each component, and how each component is connected to the others to form a pipeline.
- Describe how your code was distributed among the different components in your solution. How many lines of code were needed for the different components? Why was the code needed for each component?
- Provide example code segments from your project. Good examples would include (but are not limited to) code written for data cleaning, data storage, queries, visualization, and connecting components.
- Include a demo of the code's functionality. If you think a demo is not appropriate for your project, see your instructor.
- Report what went well and what did not go well. What would you do differently?
- Be uploaded to OneDrive and shared with your instructor.

PROPOSAL

The purpose of the proposal is to help 1) get the project organized at the start and 2) ensure that projects are unique so that there is not significant overlap. Your grade will be based on the completion of the items in the advice section and should not be interpreted as an indication of the overall project grade.

Instructions about how to submit the proposal will be given at a later time.